# PHYS 1110 Lab 2: Measuring g with a Pendulum

Time ten full swings for each string length, then divide by ten to get the period. Record every value with its unit, and keep the raw times even when a trial looks wrong. The acceleration of gravity follows from the period and the length of the string.

*Table 1. Pendulum measurements from the demonstration*

| Length (m) | Ten swings (s) | Period (s) | g (m/s$^2$) |
|---|---|---|---|
| 0.20 | 8.98 | 0.898 | 9.79 |
| 0.40 | 12.69 | 1.269 | 9.81 |
| 0.60 | 15.55 | 1.555 | 9.80 |
| 0.80 | 17.94 | 1.794 | 9.81 |

The four values agree to within a quarter of a per cent. Most of the spread comes from the moment the stopwatch is started and stopped, not from the length of the string.

*Table 2. Record your data here*

| Trial | Length (m) | Ten swings (s) | Period (s) |
|---|---|---|---|
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## Questions

1. A pendulum swings ten times in 12.7 s. What is its period?

a) 0.79 s

b) 1.27 s

c) 2.54 s

Name: _____ Section: _____ Date: _____

## Key terms

| | |
|---|---|
| **Period** | The time the pendulum takes to swing out and come back to where it started, measured in seconds. |
| **Amplitude** | How far the bob moves away from its resting point; keep it small so the period does not depend on it. |
| **Uncertainty** | An estimate of how far the measured value could be from the true one, written with the same unit as the value. |